

SHASANKA SHEKHAR PADHI

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📍 Bengaluru

Senior Scientist

Summary

A Computational scientist with around 4 years of experience in AI/ML model development, biomarker identification, and knowledge graph applications. Skilled in **GenAI, LLM, knowledge graph analytics and core bioinformatics solutions** to drive data driven insights, enhance decision making, and deliver scalable solutions through cross functional collaboration across diverse domains. Focused on advancing computational research that integrates AI and data driven science to solve complex problems across scientific domains.

Technical skills

Programming languages: Python | R | Cypher | Shell Scripting
AI/ML Frameworks: PyTorch | Scikit-learn | TensorFlow | SciPy | LLM | GenAI | GraphRAG
Deployment: Docker | GIT | LangChain
Databases: Neo4J | ChromaDB
Multimomics: Transcriptomics (RNA Seq and scRNA Seq) | Genomics (WGS)

Work Experiences

Senior Scientist

Jun '23 - Present

Syngene

Bengaluru

Received "SPOT award" for advancing bioinformatics workflow & "Performance Excellence award" for commitment and ownership in ensuring success of a critical client collaboration.

- **AI driven antibody sequence generation:** Developed an **AI driven** pipeline for antibody sequence generation, leveraging deep learning and generative models trained on large scale OAS & SAbDab antibody datasets. Integrated with immunogenicity detection, Ab-epitope docking and developability assessment to design high affinity, **developable antibodies** with therapeutic potential.
- **Codon optimization:** Developed a codon optimization workflow with **multi stage optimization**, integrating optimized sequence generation using **CodonTransformer** fine tuned on lab tested sequences and a multi objective optimization step to further optimize and rank the generated sequences using metrics like CAI, GC%, minimum free energy (mRNA), restriction enzyme site constraints among few others.
- **Knowledge Graph:** Built a KG with open source biological/RWE/clinical data, harmonized with **controlled vocabularies** for each entity. Application included drug repurposing, target identification, safety assessment for toxicity and organ wise stratification, reducing months of work to weeks.
- **Machine Learning models:** Developed an **automated ML** pipeline to build Quantitative Structure-Property Relationship (**QSPR**) model for drug property prediction, that helped reducing dependency on data scientist for model building and increased capabilities across departments.
- **Machine Learning driven biomarker identification:** Developed a computational pipeline for automated biomarker identification using **TCGA** database, by training classical **ML** models using **RNA seq** data, to identify diagnostic and prognostic signatures in BRCA, LIHC, LUAD and PRAD. Enabling faster patient stratification for precision medicine applications.
- **GWAS - AllOfUs cohort:** Developed a Cromwell (WDL) pipeline to scale up association studies across 3000+ traits using **EHRs** in the cohort to identify **statistically significant** genetic biomarkers.
- **Identifying hepatotoxic signatures and analysing their reversal in a recovery state through DEG and pathway enrichment analysis:** All the samples went through the same DEG and pathway enrichment analysis, picking genes with significantly regulated expression and assessing directionality of pathways associated with them. This helped us in pointing out how the drug helped in the recovery through monitoring the pathways that have flipped their regulation, also which class of drug toxicity does the recovery drug helps treating better.
- **Network analysis - a systems approach for target identification and prioritization**
 1. High throughput drug combination (synergy) assessment using network proximity analysis
 2. Network vulnerability analysis for target prioritization
- **In-silico KO/perturbation:** Developed a high throughput **Boolean model simulation** pipeline for in silico gene knockout/perturbation experiments, using RNA seq data to initialize the system states supporting data driven therapeutics which enhances precision in target prioritization.
- **Structure based druggability:** Developed a structure based druggability prediction pipeline leveraging parallel processing to accelerate searches across a database of known **binding pockets**, enabling rapid identification of similar sites to assess protein druggability.
- **Spectral deconvolution and compound detection:** Developed an automated compound identification pipeline leveraging **spectral data**, matched against a processed in house reference compound database.

Project Associate

Oct '22 - Jun '23

Centre for Brain Research, IISc

Bengaluru

Performed quality control, variant association and cross population studies using **GenomeIndia** GWAS data.

Personal projects

- **TrialGraph**: A Clinical Trials Knowledge Graph with an **LLM orchestrated analytical pipeline** built on Neo4j that transforms ClinicalTrials.gov data into a queryable knowledge graph, powered by a **GraphRAG** pipeline for grounded clinical Q&A.
- **CAMDA challenge**: Constructed a **Temporal Knowledge Graph** from diabetes patient records (**EHR**) using Neo4j, integrated with llama3 to setup a GraphRAG workflow for **AI driven medical applications**.
- **Biomedical research assistant**: Developed a biomedical research assistant that streamlines literature exploration using **llama3** with a **chatbot** for natural language Q&A.
- **Autism Spectrum Disorder (ASD) classification** from caregiver written behavioural text: Evaluating **lexical** and **semantic** representations for autism detection from caregiver remarks using **natural language processing** (NLP).
- **NeurIPS Open Polymer Prediction Challenge**: A Machine Learning Approach to Polymer Property Prediction: Interpretable Models. Predictive models for Tc, Tg, Rg, FFV and density of polymers.
- Built a **Deep learning** model (with **PyTorch**) using protein sequence embeddings for protein classification

Education

Master of Technology - Bioinformatics	
University of Hyderabad	Hyderabad
9.05 CGPA - GATE 2022 (AIR : 309)	
Bachelor of Engineering - Biotechnology	
Birla Institute of Technology	Ranchi
7.44 CGPA - GATE 2020 (AIR : 767)	